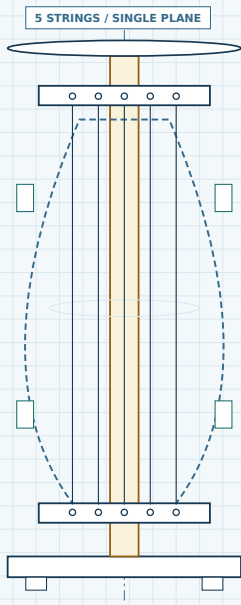


DRAWING SET AND GENERAL NOTES

PASSIVE ACOUSTIC PROTOTYPE / MEASUREMENT PLATFORM

01 / ASSEMBLY REFERENCE



Shell outline is symbolic; actual profile is unmeasured.

17 sheets. One traceable Rev B set.

Elevations, sections, part interfaces, force diagrams, candidate string calculations, and field records. This issue expands the drawing detail and corrects the supplied PDF calculations; it records no fabricated or tested assembly.

N NOMINAL / FROM REV B

C CALCULATED / CONDITIONAL

U UNRESOLVED

G01	Source record and corrections	D05	Anchor bar pattern and bending axes
A01	General arrangement: elevations	D06	Vessel support and removable top
A02	Plan sections and vessel clearance	M01	Candidate string schedule
A03	Exploded assembly and parts register	M02	Load cases and calculation record
D01	Base, feet and support geometry	M03	Stability and restraint geometry
D02	Spine section and force system	Q01	Field measurement sheets
D03	Upper anchor and tuner interface	Q02	Assembly and inspection sequence
D04	Lower anchor and frame closure	Q03	Acoustic measurement and release record

DRAWING RULE

All dimensions retain their status. N values are planning dimensions, C values depend on written assumptions, and U values require measurement, selection or calculation. Blank fields mean unresolved.

ISSUE STATUS

The primary string load path is the frame and its joints. The vessel has independent padded support. No certified capacity, final hole schedule, deflection result or acoustic performance is established by this set.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS

PROVISIONAL - NOT A CUTTING DRAWING

Actual dimensions, string data and load-path review required before fabrication or tensioning.

ISSUE 2026-09-08 | ANSI B 17 x 11 in

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NTS - DO NOT SCALE DRAWINGS

FSR-B-G00

SHEET 01 / 17 | REVIEW / MEASURE

SOURCE RECORD AND CORRECTIONS

SOURCE LINEAGE / CORRECTIONS APPLIED TO THIS DRAWING EXPANSION

SOURCE REGISTER

ID	SUPPLIED SOURCE	USE IN THIS ISSUE
S1	Rev B Schematic Update (DOCX), dated 5 Sep 2026	Nominal geometry, parts P1-P15, independent vessel support and release conditions.
S2	String tension gate PDF, 15 pages	Worked examples and CX-1 proposals. Recomputed; no part rating accepted from narrative.
S3	Updated schematic PNG, two attached copies	Byte-identical reference drawing. One source, not two confirmations.
S4	1000012794.png + GitHub screenshot	Appearance reference only. Collar shapes and depicted hardware are not fabrication evidence.
S5	Five_String_Acoustic_Resonator_Rev_B.md	Conversation transcript despite filename. Not the original design specification.

REFERENCE SOURCES CHECKED 8 SEP 2026

W1 UNSW - ideal string relation

W2 Baker & Haynes - static equivalence

W3 USDA Forest Products Laboratory - Wood Handbook

W4 D'Addario - current string tension calculator

W4: the historical chart URL redirected to the current calculator. The attached unit weights are transcribed inputs, not independently reconfirmed product data. The source archive records filenames and SHA-256 hashes.

CALCULATION CORRECTIONS / S2

PAGE	CORRECTION / CONSEQUENCE
p5	G3 is 195.9977 Hz at A4 = 440 Hz equal temperament. 195.00 was a label error. Recomputed total: 70.84 lbf.
p7	Rectangular kern: S/A = d/6, not d/2. Moving the string-plane force to the face still requires its equivalent couple.
p8-10	Attachment span is not automatically speaking length. Clear-wood averages and Euler examples do not rate an ungraded member or identify the first failure.
p11	A couple uses perpendicular reaction separation. The collar estimate $H = Pe/d$ lacks a resolved reaction geometry.
p12	For the stated two-string cantilever at a 1.5 in-wide support, M is about 69.09 lbf-in on the left, not 113.
p12-15	No universal 11 in bar minimum or 2D rule is adopted. A 1.5 in-high bar cannot contain bolt centers 1.5 in apart with positive top/bottom margins.
p13-15	Bolt gauge runs along z, not across the 2.5 in section depth. M/g is a couple force, not a complete per-bolt prying check.

Audit additions on A02 and D05 are conditional geometry checks. This set does not select CX-1 as an approved connection. The original source files remain the historical record.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
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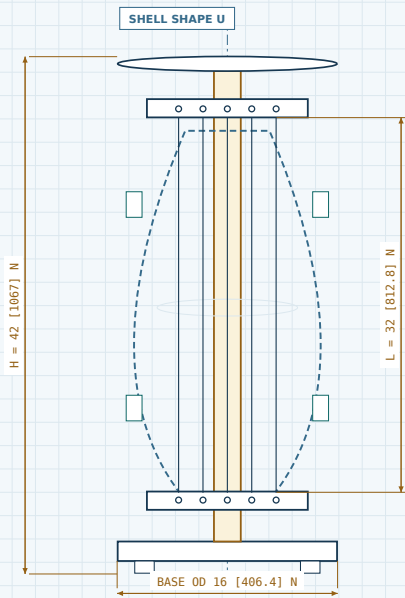
FSR-B-G01

SHEET 02 / 17 | REVIEW / MEASURE

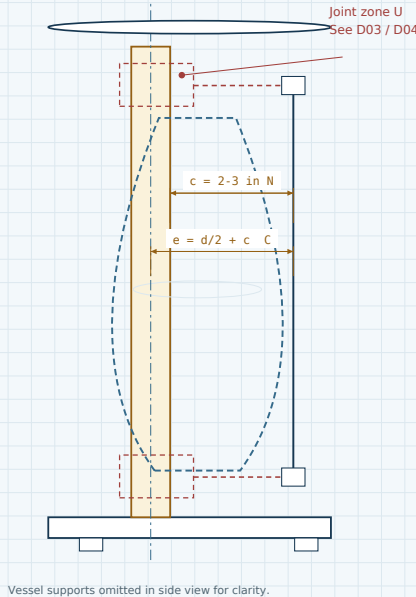
GENERAL ARRANGEMENT: ELEVATIONS

ALIGNED ORTHOGRAPHIC VIEWS / ALL OUTLINES NTS

01 / FRONT ELEVATION



02 / SIDE ELEVATION (+y TO RIGHT)



DATUM / DIMENSION REGISTER

ITEM	VALUE / STATUS
A	Base upper face: field datum z = 0.
B	Spine centroid axis: x = y = 0.
x	Across the five-string array.
y	From spine axis toward strings.
z	Vertical, positive upward.
L	32 in N; planning 28-36. Bearing-to-bearing; five lengths within 1/16 in unless scaled.
H	42 in N; planning 36-48. Overall datum chain U.
d, b	Actual spine depth and width U.
c	Face-to-string-plane gap; min 1.5 in in S1.

Vertical closure is unresolved: nominal 40 in spine, base thickness, feet, top mount and overlaps must be reconciled with nominal 42 in overall height. Do not derive cutting lengths from this elevation.

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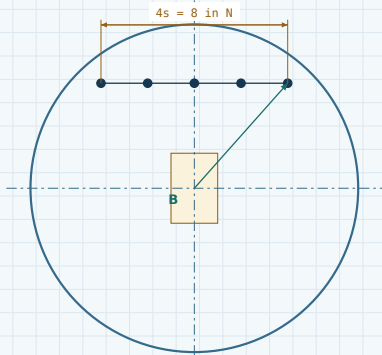
FSR-B-A01

SHEET 03 / 17 | REVIEW / MEASURE

PLAN SECTIONS AND VESSEL CLEARANCE

FIT THE COMPLETE HARDWARE ENVELOPE AT EVERY OCCUPIED HEIGHT z

01 / TRANSVERSE SECTION THROUGH STRINGS



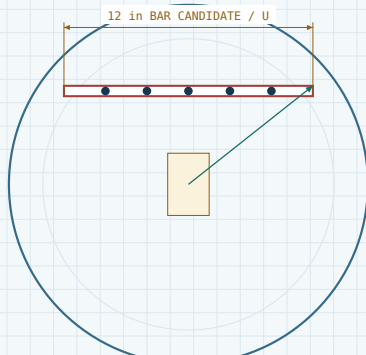
Example: d = 3, c = 3, s = 2, radial allowance = 1 in.

$D_{clear}(z) \geq 2 \left[\sqrt{x_{max}^2 + y_{max}^2} + allowance(z) \right]$

Centered circular section only. For the strings alone: $2[\sqrt{4^2 + 4.5^2} + 1] = 14.04$ in. This is a conditional screen, not a measured vessel diameter.

General case: measure the actual inner boundary at each z and check all hardware corners, string excursion, tool access and installation path. A mouth smaller than the assembly can prevent insertion even if the belly fits.

02 / CONDITIONAL CX-1 BAR ENVELOPE



Bar thickness assumed 0.5 in along y; no tuners or bolt heads.

For the illustrated 12 in bar: $2[\sqrt{6^2 + 4.75^2} + 1] = 17.31$ in before additional hardware. A 14.04 in string-only envelope cannot prove that this bar fits.

Required decision: keep anchors outside the vessel, revise the hardware envelope, change the vessel, or revise the overall geometry with review. Anchor elevations relative to the mouth remain U.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

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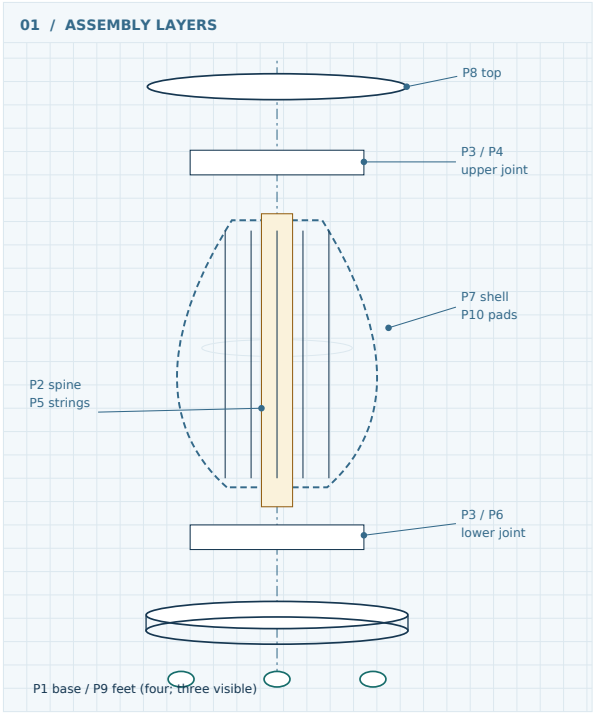
FSR-B-A02

SHEET 04 / 17 | REVIEW / MEASURE

EXPLODED ASSEMBLY AND PARTS REGISTER

PART IDENTITIES FOLLOW S1 / EXPLODED SPACING DOES NOT SPECIFY ASSEMBLY HEIGHTS

01 / ASSEMBLY LAYERS



ID	PART	QTY	SPECIFICATION / STATUS
P1	Base	1	Two 3/4 in laminations; 16 in OD N. D01.
P2	Spine	1	Hard maple/ash; nominal 2 x 3 x 40 in. Actual U.
P3	Anchor bars	2	All section, holes and connections subject to review.
P4	Tuners	5	Exact product and mounting drawing U. D03.
P5	Strings	5	Candidate calculation M01; selected set U.
P6	Nut / bridge	2	Replaceable rounded bearing strips; shape U.
P7	Vessel	1	Material, profile, mass, wall and mouth U; measure per D06.
P8	Top plate	1	Thin metal or 1/4-3/8 in plywood; 14-18 in OD; mount U.
P9	Feet	4	Neoprene/Sorbothane, 1-1.5 in OD N; equal height; through-bolted.
P10	Padded supports	3-4	Straps/cleats independent of the string force loop.
P11	Restraint	1	Geometry and capacity determined separately. M03.
P12	Guards	As needed	Polycarbonate, anchor/tuner access and clearance U.
P13	Fasteners	Set	Grade-marked bolts, washers, locking nuts; size U. No wood screws in primary tension path.
P14	Passive pickup	Optional	Isolated, removable; mounting recorded. Q03.
P15	Powered exciter	Phase 2	Excluded from passive baseline. S1 constraints apply.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

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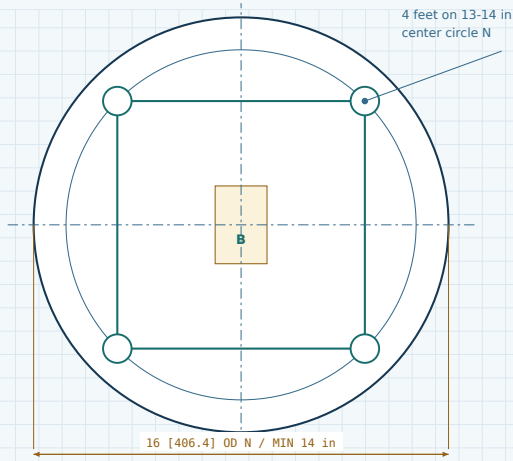
FSR-B-A03

SHEET 05 / 17 | REVIEW / MEASURE

BASE, FEET AND SUPPORT GEOMETRY

BASE DIMENSIONS N / FASTENER SIZES AND HOLE LOCATIONS U

01 / BASE PLAN



PRIMARY FRAME ATTACHMENT

Locate the spine shoe, backing structure and any braces from the reviewed frame joint. The circular base outline does not specify bolt centers. Include base bending, local bearing, pull-through and joint rotation.

02 / BASE EDGE SECTION

Two nominal 3/4 in Baltic-birch layers (P1)



Bond specification, clamp method, cure and through-bolt details: U. Feet shown symbolically; actual compressed heights govern level and datum A.

SUPPORT GEOMETRY / CENTERED POINT-FOOT EXAMPLE

$$a = D_{\text{foot}} / (2 \sqrt{2})$$

For $D_{\text{foot}} = 13-14$ in, the square side-to-center distance is 4.60-4.95 in. This is smaller than the base radius. Use actual contact patches and loaded compliance for the final support hull.

FIELD RECORD

Record each foot center (x, y), patch size, compressed height, fastener ID and mounting surface. Verify four contacts without rocking. Plot the loaded center-of-gravity projection on M03.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS

PROVISIONAL - NOT A CUTTING DRAWING

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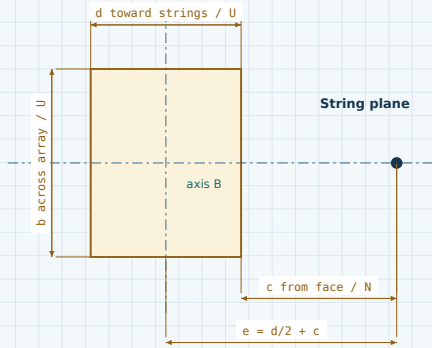
FSR-B-D01

SHEET 06 / 17 | REVIEW / MEASURE

SPINE SECTION AND FORCE SYSTEM

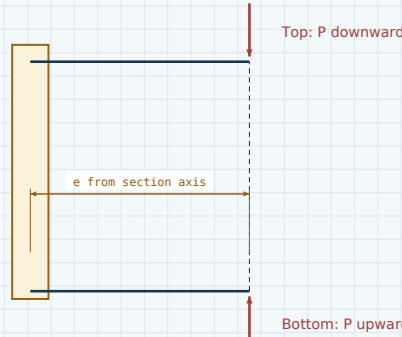
FIRST-ORDER STATIC MODEL / NO WOOD GRADE OR CONNECTION CAPACITY HAS BEEN ASSIGNED

01 / SPINE CROSS-SECTION



02 / UPPER AND LOWER ACTIONS ON THE FRAME

Paired forces close through the frame. Vessel is outside this force loop.



$$P = \sum(T_i) \quad M_x = -\sum(y_i T_i) \quad M_y = \sum(x_i T_i)$$

Signs shown for downward forces at the upper connection; lower forces reverse. A moved force requires the equivalent couple. Do not replace e with d/2 merely because a bracket touches the spine face. [W2]

$$A = b d \quad I_x = b d^3/12 \quad S_x = b d^2/6$$

$$\sigma = P/A \pm |M_x|/S_x \quad (\text{one-axis model})$$

$$S_x/A = d/6 \quad (\text{rectangular-section kern})$$

Formula gives stress demand only; no allowable stress is selected here.

Use measured net sections and both bending axes when needed. Holes, defects, moisture, grade, bearing, restraint, creep and connection rotation remain part of review. Full 2 x 3 in and dressed 1.5 x 2.5 in sections are examples, not interchangeable stock.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS

PROVISIONAL - NOT A CUTTING DRAWING

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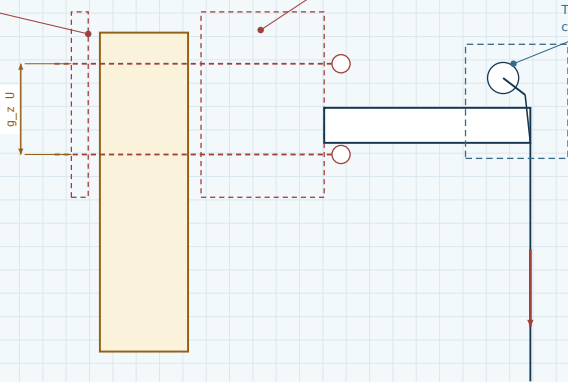
FSR-B-D02

SHEET 07 / 17 | REVIEW / MEASURE

UPPER ANCHOR AND TUNER INTERFACE

UPPER JOINT CONCEPT / FASTENER CENTERS, SIZES AND REACTION GEOMETRY UNRESOLVED

01 / UPPER HEAD-BLOCK INTERFACE, SIDE SECTION

Back plate footprint U
Full bearing / accessHead block / shoulder U
Positive vertical reactionTuner winding
clearance U

Dashed red components reserve a review zone; they are not fabrication geometry.

INTERFACE REGISTER

ID	REQUIRED INPUT
U01	Tuner model, string end type, mounting thickness and fastener drawing.
U02	String run from winding post to nut; break angle and individual support reactions.
U03	Head block and shoulder geometry, grain, contact area and slip restraint.
U04	Bolt diameter, gauge along z, hole fits, end/edge margins and backing plate.
U05	Coupled bolt tension/shear, prying, bearing, splitting, net section and rotation.
U06	Guard, winding tool and string replacement swept envelopes.

FORCE RESOLUTION AT A BEARING

At a nut or bridge, use the vector sum of the tensions in the adjacent string segments. The bearing reaction is not automatically equal to one string tension. The total upper subassembly still transfers the full string resultant into the frame.

DETAIL STATUS

The PDF CX-1 hardware remains a candidate. No drill diameter, torque, metal grade substitution or operating load is released on this sheet.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS

PROVISIONAL - NOT A CUTTING DRAWING

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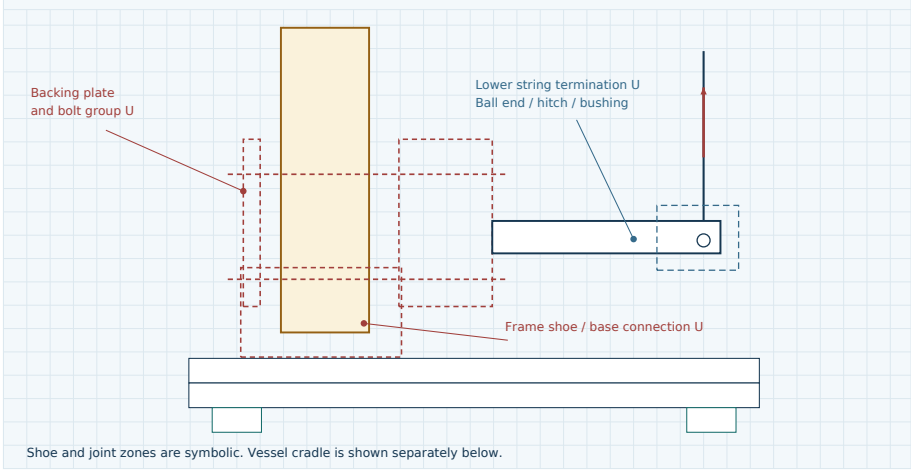
FSR-B-D03

SHEET 08 / 17 | REVIEW / MEASURE

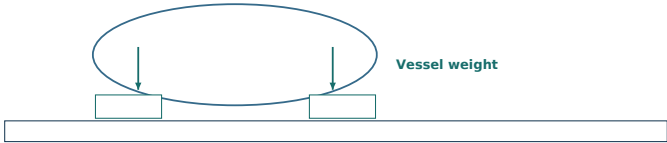
LOWER ANCHOR AND FRAME CLOSURE

LOWER JOINT CONCEPT / CLOSE STRING FORCES THROUGH THE FRAME

01 / LOWER ANCHOR, SIDE SECTION



02 / INDEPENDENT VESSEL SUPPORT CONCEPT



LOWER END IS A SEPARATE REVIEW

L01 TERMINATION

Select compatible ball-end, loop or hitch geometry. Resolve local bearing, pull-through, tear-out, bend radius and broken-end containment.

L02 LOWER JOINT

Review the upward string force, eccentric couple and transverse bar bending. An inverted upper sketch does not prove identical contact or prying behavior.

L03 BASE CLOSURE

Include the actual shoe, backing members and plate-to-base load transfer. Avoid counting the full internal string force as additional external weight.

L04 RESTRAINT

Review anti-tip anchorage under its own external load cases. It is not automatically provided by the string-anchor bolts.

The vessel is supported for its own weight and motion using padded contact. No part of the shell, neck or rim is a string-force reaction member. Support-pad sizes and preload remain U.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

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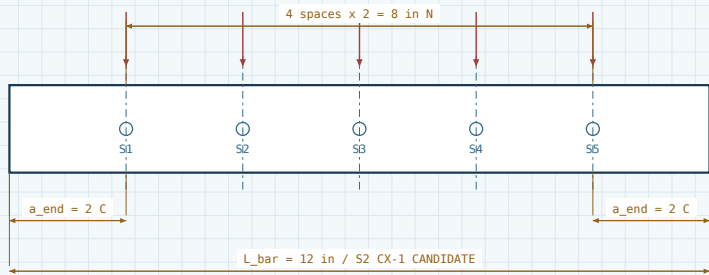
FSR-B-D04

SHEET 09 / 17 | REVIEW / MEASURE

ANCHOR BAR PATTERN AND BENDING AXES

STRING CENTERLINE LAYOUT / HOLE AND HARDWARE DETAILS NOT RELEASED

01 / BAR FRONT ELEVATION (x-z)



Centerlines only. s = 2 in N; planning range 1.5-2.25 in. All drilling and terminations U.

GEOMETRY AND BENDING CHECKS

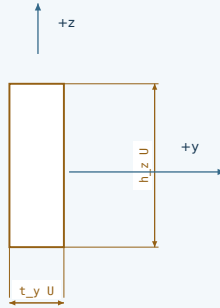
$$L_{\text{bar}} = (N - 1)s + 2 a_{\text{end}} = 8 + 2 a_{\text{end}}$$

Neither 10, 11 nor 12 in is a universal required length. End allowance follows the actual termination, hole, material, load direction and review. A 12 in bar leaves 2 in at each end geometrically; that does not establish a capacity.

$$M_{\text{left}} = 16.2525(4 - .75) + 13.0187(2 - .75) \\ = 69.09 \text{ lbf-in } C \quad (S2 \text{ states } 113)$$

This example assumes continuous support across a centered 1.5 in width and vertical string forces. Right side: 68.47 lbf-in. Change the support or force direction and recalculate. No net-section stress or deflection result is certified.

02 / SECTION (y-z)



Section orientation must be declared.

$$S_y = t_y h_z^2 / 6$$

For vertical loads on an x-spanning bar, h_z is the bending depth. For the illustrated candidate t_y = 0.5 and h_z = 1.5 in: S_y = 0.1875 in^3 before holes.

A 1.5 in-high bar cannot contain two z-spaced bolt centers 1.5 in apart with positive margins. A separately sized connection block or revised geometry is required for that proposal.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

Actual dimensions, string data and load-path review required before fabrication or tensioning.

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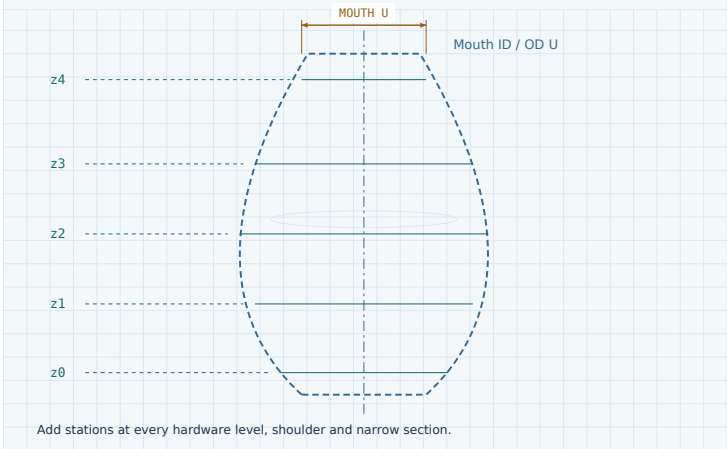
FSR-B-D05

SHEET 10 / 17 | REVIEW / MEASURE

VESSEL SUPPORT AND REMOVABLE TOP

INDEPENDENT SHELL SUPPORT / AIR GAP AND ASSEMBLY PATH MUST BE MEASURED

01 / PROFILE STATIONS



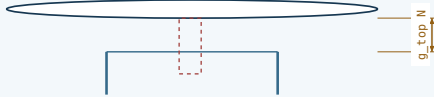
FIT RECORD

Measure wall thickness, inside boundary, mouth, base, material, mass and defects. Record the occupied hardware envelope and required movement allowance at each station. Do not infer inside diameter from outside appearance.

Cavity volume remains undetermined: $V = \text{integral } A(z) dz$ requires measured internal areas and integration limits. Opening shape, gaps and acoustic coupling are also unmeasured; no Helmholtz frequency or resonant performance is assigned.

02 / TOP PLATE AND ANNULAR GAP

Top diameter 16 in N; permitted planning range 14-18 in.



Gap $g_{top} = 1.5$ in N, planning range 1-2 in. Captive removable mount U. Keep shell and plate from unintended hard contact.

03 / PADDED CONTACT DETAIL



SUPPORT / ACCESS RECORD

Pad area, preload, attachment method and removal path are U. Avoid point loads on glass; never drill tempered glass. Confirm guard and tuner access without contacting the shell. [S1 sections 4-5]

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

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FSR-B-D06

SHEET 11 / 17 | REVIEW / MEASURE

CANDIDATE STRING SCHEDULE

EX-B1-R1 / RECOMPUTED CANDIDATE ONLY / NO STRING PURCHASE OR OPERATING SET IS RELEASED

Common assumptions: L = 32.00 in (0.8128 m), A4 = 440 Hz equal temperament, flexible-string model, source-PDF unit weights. Product names and aliases are carried from S2; their current equivalence and usable string length require maker confirmation.

ID	NOTE	f / Hz	SOURCE PRODUCT LABEL	UW / lb/in	mu / g/m	T / lbf	T / N
S1	A2	110.0000	NW026 / .026 wound	0.00012671	2.2628	16.25	72.29
S2	C3	130.8128	LE018 / PL018 / .018 plain	0.00007177	1.2817	13.02	57.91
S3	E3	164.8138	LE014 / PL014 / .014 plain	0.00004342	0.7754	12.50	55.61
S4	G3	195.9977	LE012 / PL012 / .012 plain	0.00003190	0.5697	12.99	57.78
S5	C4	261.6256	LE010 / PL010 / .010 plain	0.00002215	0.3956	16.07	71.49

RECOMPUTED TOTAL: 70.84 lbf / 315.09 N

SOURCE PDF: 70.8 lbf rounded

T = mu (2Lf)^2

T_lbf = UW (2 L_in f)^2 / 386.4

1 lbf = 4.448221615 N

The inherited 386.4 convention is used to reproduce the maker-style calculation in S2. Calculation precision does not increase confidence in the transcribed UW. [S2 pp1-5; W1]

DATA THAT STILL NEED CONFIRMATION

Exact product, construction, unit weight, total length, winding/post fit, termination, usable pitch range and individual reviewed limit. Tuner and joint review must cover the actual string path.

20 lbf/string and 100 lbf total remain planning ceilings. A calculated candidate below them is not an authorized operating set. Use the lower reviewed operating limits; missing data leave the assembly untensioned.

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

Actual dimensions, string data and load-path review required before fabrication or tensioning.

ISSUE 2026-09-08 | ANSI B 17 x 11 in
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NTS - DO NOT SCALE DRAWINGS

FSR-B-M01

SHEET 12 / 17 | REVIEW / MEASURE

LOAD CASES AND CALCULATION RECORD

DEMAND CALCULATIONS ARE CONDITIONAL / ALL CAPACITY AND DEFLECTION FIELDS REMAIN U

CASE REGISTER

CASE	CONFIGURATION	CHECK
LC0	Untensioned assembly	Self-weight, handling, shell supports, feet and restraint installation.
LC1	All five at reviewed targets	Use each T_i, x_i and y_i; axial force and both bending moments.
LC2	Staged tuning / one-sided loading	Check maximum imbalance during the declared tuning sequence.
LC3	One string slack or broken	Residual static forces plus separately reviewed release/transient effects.
LC4	Tuning torque / handling	Resolve tuner torque, operator forces and accessory loads.
LC5	Tip / slip / restraint	External loads, mass, CG and restraint geometry. See M03.

CONDITIONAL NUMERIC SCREEN

P = 70.8354 lbf

e = 2.5/2 + 3 = 4.25 in

|M_x| = 301.05 lbf-in

sum(x_i T_i) = -0.78 lbf-in, for x = -4, -2, 0, +2, +4 in

Illustrative actual section b = 1.5, d = 2.5 in: A = 3.75 in^2; S_x = 1.5625 in^3. First-order one-axis front stress = 211.56 psi; back stress = 173.78 psi tension.

This is a demand example, not a rating for a nominal 2 x 3 or a drilled joint.

COMPLETE BEFORE A FABRICATION OR TENSIONING RELEASE

ITEM	MEASURE / DEFINE	REVIEW METHOD	LIMIT / RESULT	EVIDENCE ID
Spine / frame	Actual b, d, span, defects	Net section; stability; creep	U / U	U
Bars / head blocks	Material, orientation, holes	Bending; shear; deflection	U / U	U
Connections	Bolts, bearing faces, gauge	Combined actions; slip; splitting	U / U	U
Deflection / clearance	Initial shape; joint motion	Loaded motion + residual set	U / U	U
Strings / tuners	Product and installation	Target, safe range, local forces	U / U	U

PUBLIC DOMAIN - CC0 1.0 | AUTHOR: ANONYMOUS
PROVISIONAL - NOT A CUTTING DRAWING

Actual dimensions, string data and load-path review required before fabrication or tensioning.

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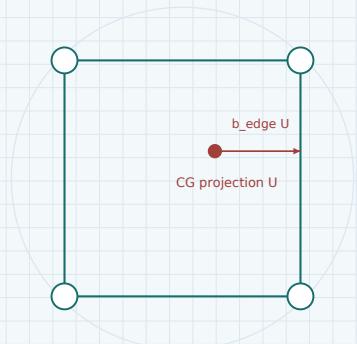
FSR-B-M02

SHEET 13 / 17 | REVIEW / MEASURE

STABILITY AND RESTRAINT GEOMETRY

EXTERNAL EQUILIBRIUM / STRING TENSION IS INTERNAL TO THE CLOSED ASSEMBLY

01 / PLAN: ACTUAL FOOT-CONTACT HULL



Find the minimum distance to every relevant pivot edge.

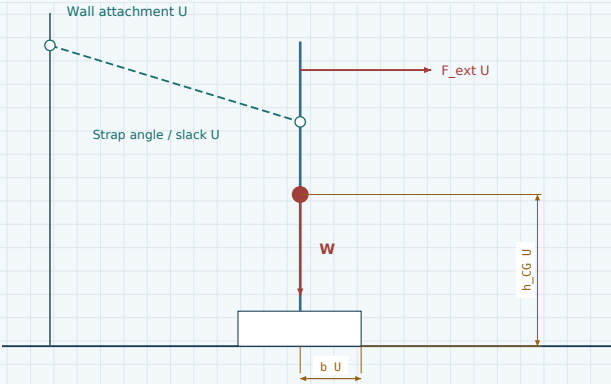
STATIC SCREEN / NO SLIDING / NO RESTRAINT CONTRIBUTION

$F_{ext} h_{load} < W b_{edge}$

$\tan(\theta_{tip}) = b_{edge} / h_{CG}$

Use the actual direction of loading and the nearest relevant pivot edge. The old 10 degree tilt is a calculation example only; it is not an instruction to tilt the apparatus.

02 / SIDE: WEIGHT, LOAD AND RESTRAINT



RESTRAINT DEMAND

Calculate using the actual attachment coordinates, allowable load directions, wall/substrate anchorage, slack and engagement path. Dynamic engagement cannot be inferred from a static label or a generic 200 lbf rating.

Record mass and CG in x, y and z, foot contact behavior, external test load, angle, force, fixture and abort condition. Slip and tip require separate checks. No anti-tip rating is assigned here.

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PROVISIONAL - NOT A CUTTING DRAWING

Actual dimensions, string data and load-path review required before fabrication or tensioning.

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FSR-B-M03

SHEET 14 / 17 | REVIEW / MEASURE

FIELD MEASUREMENT SHEETS

BLANK RECORDS ARE U / COMPLETE WITH UNITS, UNCERTAINTY AND EVIDENCE

CONFIGURATION ID: DATE: MEASURED BY:

01 / VESSEL AND OCCUPIED ENVELOPE

STATION	z FROM A	INNER PROFILE / ID	HARDWARE ENVELOPE	MIN CLEARANCE / MOTION	PHOTO / METHOD
Mouth / entry					
Upper anchor					
Upper shoulder					
Maximum belly					
Lower shoulder					
Lower anchor					
Support / base					

02 / DATUM AND STOCK CLOSURE

ITEM	ACTUAL / UNITS	EVIDENCE
Spine b x d / length		
Section orientation / grain		
Base + feet stack height		
Upper / lower bearing z		
Overall H / top gap		

03 / CONNECTION AND SUPPORT INPUTS

ITEM	ACTUAL / UNITS	EVIDENCE
Bar / block / backing sizes		
Hole axes / diameter / margins		
Bolt group gauge / material		
Foot coordinates / patches		
Mass / CG / restraint geometry		

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FSR-B-Q01

SHEET 15 / 17 | REVIEW / MEASURE

ASSEMBLY AND INSPECTION SEQUENCE

SEQUENCE FROM S1 / CONDITIONAL ON DIMENSIONAL AND STRUCTURAL REVIEW

01

CLOSE THE DIMENSIONAL MODEL

Complete Q01. Check vessel entry path, all occupied sections, vertical stack and service access. Retain N values only if the measured arrangement supports them.

02

COMPLETE THE STRUCTURAL REVIEW

Select actual members, connections, strings and reviewed operating limits. Resolve D03-D05 and M02-M03; issue a separate cutting and drilling schedule.

03

DRY-FIT THE FRAME

After review, fit base, spine, head block and lower anchor without the vessel. Use the reviewed bolts, backing members and braces. Record tightening requirements.

04

FIT SUPPORTS AND PROTECTION

Establish foot contacts and reviewed restraint. Fit independent padded vessel support, nut/bridge, guards and cable strain relief. Confirm every clearance again.

05

INSTALL AND TUNE TO REVIEWED TARGETS

Use the reviewed sequence. Inspect near 25%, 50%, 75% and 100% of those targets, not automatically the planning ceiling. Stand out of the string plane; wear eye protection.

06

OBSERVE AND RECORD

After load review, observe at operating tension for 30 min; measure loaded deflection and unloaded recovery. Compare with the predeclared limits. This is not proof-load certification.

07

FIT THE REMOVABLE TOP AND MEASURE

Fit the documented captive mount once frame stability is established. Record the actual coupling and proceed to the passive acoustic baseline on Q03.

STOP / ABORT

Cracking, progressive movement, slip, permanent deformation, unstable footing, loss of clearance or vessel contact. Remove load using the reviewed safe method; record the condition and reassess the affected joint.

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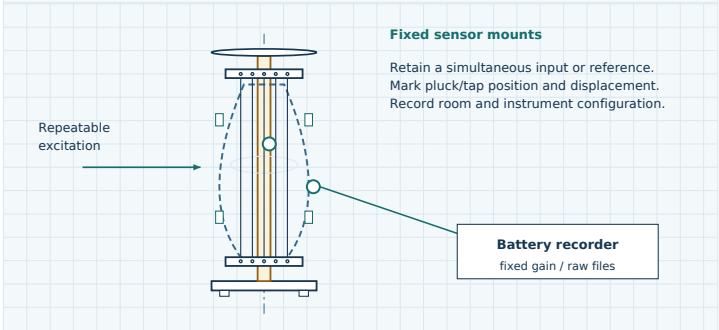
FSR-B-Q02

SHEET 16 / 17 | REVIEW / MEASURE

ACOUSTIC MEASUREMENT AND RELEASE RECORD

PASSIVE BASELINE / COMPLETE RECORD BEFORE CLAIMING A MEASURED RESULT

01 / REPEATABLE MEASUREMENT SETUP



$Q = (2 \pi / \ln(10)) f_0 t_{10dB}$

Single isolated mode with exponential amplitude decay only. Record fit band, interval, noise floor and uncertainty. Unresolved beating, clipping, gain changes or multiple modes prevent a single-mode Q estimate.

Keep raw recordings and configuration ID, date/time, string identities, pitches/tensions, sample rate, gain, sensors, excitation, room state, temperature, anomalies and stop reason. Start with at least three repeats; vary one factor per comparison.

Phase 2 source limit: isolated fused 12-24 VDC, <=30 W, no mains inside assembly; mount temperature rise <=10 C after 10 min. No powered exciter is part of this baseline.

MEASUREMENT REGISTER / S1 T0-T8

TEST	ACTION	RECORD / INTERPRETATION
T0-T2	Inspect / observe / assess	Complete reviewed mechanical and stability criteria first.
T3	Component tap map	Damp strings; fixed mounts and repeatable excitation.
T4	Individual strings	Pitch, harmonics, drift and defined 10 dB decay interval.
T5	Transfer response	H = Y/F only with measured input; otherwise response ratio.
T6	Airflow map	Tufts clear of strings; optional guarded external fan >=12 in away.
T7	Room sweep	External speaker >=3 ft away; 20-200 Hz; predeclared SPL ceiling.
T8	Phase 2 only	After T0-T7 pass; separate reassessment and S1 electrical limits.

COMPLETION RECORD / UNEXECUTED

RELEASE ITEM	STATUS	REVIEW / DATE
Measured geometry	UNRESOLVED	
String + connection limits	UNRESOLVED	
Stability + restraint	UNRESOLVED	
Fabrication / load / acoustics	NOT EXECUTED	

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SHEET 17 / 17 | REVIEW / MEASURE